26: Poisson Regression and Overdispersion

Stat 230 - Spring 2026

1 Data Overview

These data come from a study of 151 separate 3 hectare sites in Victoria, Australia. The purpose of the study was to learn what makes a good habitat for possums, in order to identify areas of forest for habitat preservation. The variables are:

- y, the number of possum species found on the site
- `Acacia`, basal area of acacia (a shrub), in m^2
- `Bark`, a bark quality index (bigger is better)
- `Habitat`, a habitat score for Leadbeater's possums (bigger is better)
- `Shrubs`, the number of shrubs
- `Stags`, the number of hollow trees
- `Stumps`, indicator for whether or not stumps from logging are present

Load the dataset:

```
possums <- read.csv("https://aluby.github.io/stat230/data/possums.csv", header = TRUE
```

2 Part 1: Is the Poisson model appropriate?

1. Fit the Poisson regression model using y as the response variable and all of the other variables as main effects. Interpret the estimated effect of the number of hollow trees (`Stags`).

```
possum_glm <- glm(___ ~ Acacia + Bark + Habitat + Shrubs + Stags + Stumps,
                  data = possums,
                  family = ___)
summary(possum_glm)
```

2. A Poisson model assumes that the variance equals the mean. We can estimate the dispersion parameter using the Pearson χ^2 statistic: $\widehat{\psi} = \frac{X^2}{n-(p+1)}$. Run the code below, then say what the value of $\widehat{\psi}$ tells you about these data.

```r
X2 <- sum(residuals(possum_glm, type = "pearson")^2)
X2
df.residual(possum_glm)

psi_hat <- X2 / df.residual(possum_glm)
psi_hat
```

3. Now fit the quasipoisson regression model with the same predictors (call it `possum_quasi`). Find the estimated dispersion parameter in the `summary()` output. How does it compare to your answer in Question 2? Argue that, since $\widehat{\psi} < 1$, using the "regular" Poisson model gives a more conservative (cautious) assessment of uncertainty than the quasipoisson model. (*Hint:* how are the two sets of standard errors related?)

4. Take a look at the fitted values (the estimated means) from the Poisson model. Should we expect the Pearson residuals to follow a normal distribution in this setting? Can we trust the Pearson χ^2 goodness-of-fit test?

```r
summary(fitted(possum_glm))
```

3 Part 2: Inference

5. Give a 95% confidence interval for the effect of the number of hollow trees on the mean number of species.

a. Use the Poisson model and `confint.default()` (remember to exponentiate!).
b. Now repeat using the quasipoisson model. Use $\hat{\beta} \pm t^* \, SE_{quasi}(\hat{\beta})$ with `qt(.975, df.residual(possum_quasi))`. Which interval is wider? Why?

6. Drop `Acacia`, `Shrubs`, and `Stumps` from your model (call the reduced model `possum_reduced`). On the basis of the drop-in-deviance (likelihood ratio) test, is there evidence against this reduced model? State the hypotheses, the test statistic and degrees of freedom, and your conclusion.

```r
possum_reduced <- glm(___ ~ ___, data = possums, family = poisson)

anova(possum_reduced, possum_glm, test = "LRT")
```

Does the answer change if you use the quasipoisson models and `test = "F"`? Why does (or doesn't) it?

7. Using the reduced model from Question 6, give an estimate of the mean number of species when the bark index is 8, the habitat score is 4, and the number of hollow trees is 10. Give a 95% confidence interval for this estimate by treating $\log(\hat{\mu})$ as approximately normally distributed. Why is this justified?

```
predict(possum_reduced, newdata = data.frame(Bark = ___, Habitat = ___, Stags = ___),
        type = "link", se.fit = TRUE)
```

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4 Function quick reference

Function	Purpose
`glm(y ~ x, data, family = poisson)`	Fit a Poisson regression model
`glm(y ~ x, data, family = quasipoisson)`	Fit a quasipoisson regression model
`residuals(model, type = "pearson")`	Pearson residuals; X^2 is the sum of their squares
`df.residual(model)`	Residual degrees of freedom, $n - (p + 1)$
`confint.default(model)`	Wald confidence intervals for the coefficients (use `exp()` to get multiplicative effects)
`anova(reduced, full, test = "LRT")`	Drop-in-deviance (likelihood ratio) test for Poisson models
`anova(reduced, full, test = "F")`	Drop-in-deviance F-test for quasipoisson models
`predict(model, newdata, type = "link", se.fit = TRUE)`	Predicted log mean and its standard error